CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CL)

November 2013

CL 301 Transportation Engineering-I

Date: 18.11.2013, Monday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Design flexible pavement for construction of a two lane single carriageway with the following data as per IRC method. [07]

Initial traffic in the year of completion of construction=400cvpd

Annual growth rate of traffic=7.5%

Design life=15 years

Vehicle damage factor=2.5

Design CBR of sub-grade=4%

Use fig: 1 & fig: 2 provided with question paper for code of practice.

Q - 2 (a) Explain alternate bay method of construction of cement concrete roads. Also, write how it defers from continuous method. [04]

(b) Calculate spacing between contraction joint for a 3.8 m wide PCC slab having thickness of 12.5 cm. Assume following data: [05]

Coefficient of friction=1.5

Unit weight of cement concrete=2360 kg/cm^3

Ultimate stress in tension in cement concrete=1.75 kg/cm^2

Factor of safety=2

(c) Discuss about highway projects administration. [05]

OR

Q - 2 (a) Determine the spacing between expansion joints in a cement concrete pavement by using following data: [05]

Concrete temperature during laying operation:15° C

Max. Slab temp. In summer=60° C

Thermal coefficient of expansion of concrete=8 X 10^{-8} / °C

Max. joint width=2.5 cm

(b) Discuss the importance of gross wheel load and contact pressure in stress distribution pattern and in pavement design. [04]

(c) Explain methods of raising highway finances. [05]

Q - 3 (a) Calculate the length of transition curve needed for highway having a longitudinal circular curve of radius 300 m in plain terrain. [06]

Design speed=65 kmph

Rate of introduction of superelevation is 1 in 150

Pavement width including extra widening=7.5 m

(b) Calculate the passing sight distance for a two way traffic highway for which the design speed is 60 kmph. The rate of acceleration of the fast moving vehicle may be assumed as 3.6 kmph per sec. Difference between the overtaking vehicle and overtaken vehicle is 20 kmph. What will be the passing sight distance if only one way traffic is allowed? [06]

(c) What is prime coat? [02]

OR

Q - 3 (a) A vertical summit curve is formed at the intersection of two gradients +3% and -5%. Design the length of summit curve to be provided a stopping sight distance for a design speed of 80 kmph. Reaction time is 2.5 sec. coefficient of friction is 0.35. [06]

(b) Explain extra widening of pavements on horizontal curve with sketch and formula. [06]

(c) What is tack coat? [02]

SECTION - II

Q - 4 (a) Describe the procedure for enoscope method. [04]

(b) What is channelized intersection? Draw neat sketch. [03]

Q - 5 (a) Write a note on Obligatory points. [04]

(b) Explain Macadam's construction with neat sketch. [05]

(c) Enlist different test to be performed on aggregate and explain any one. [05]

OR

Q – 5 (a) Explain Map study for engineering survey. [04]

(b) Calculate the road length of a district as per Nagpur road plan. [05]

(i) Total area = 6300 km^2

(ii) length of railway track = 75 km

(iii) Agricultural area = 2800 km^2

No. of villages with population ranges <500, 501-100, 1001-2000, 2001-5000 and above 5001 are 450, 320, 110, 50 and 10 respectively.

(c) Write a note on float test. [05]

Q – 6 (a) List out various flexible pavement failures. [04]

(b) Distcuss importance of highway drainage. [05]

(c) List various regulatory signs and draw any four. [05]

OR

Q – 6 (a) Give factors affecting pavement failure for alligator cracking. [04]

(b) Define Sub surface drainage and give method for lowering the water table. [05]

(c) List various warning signs and draw any four. [05]

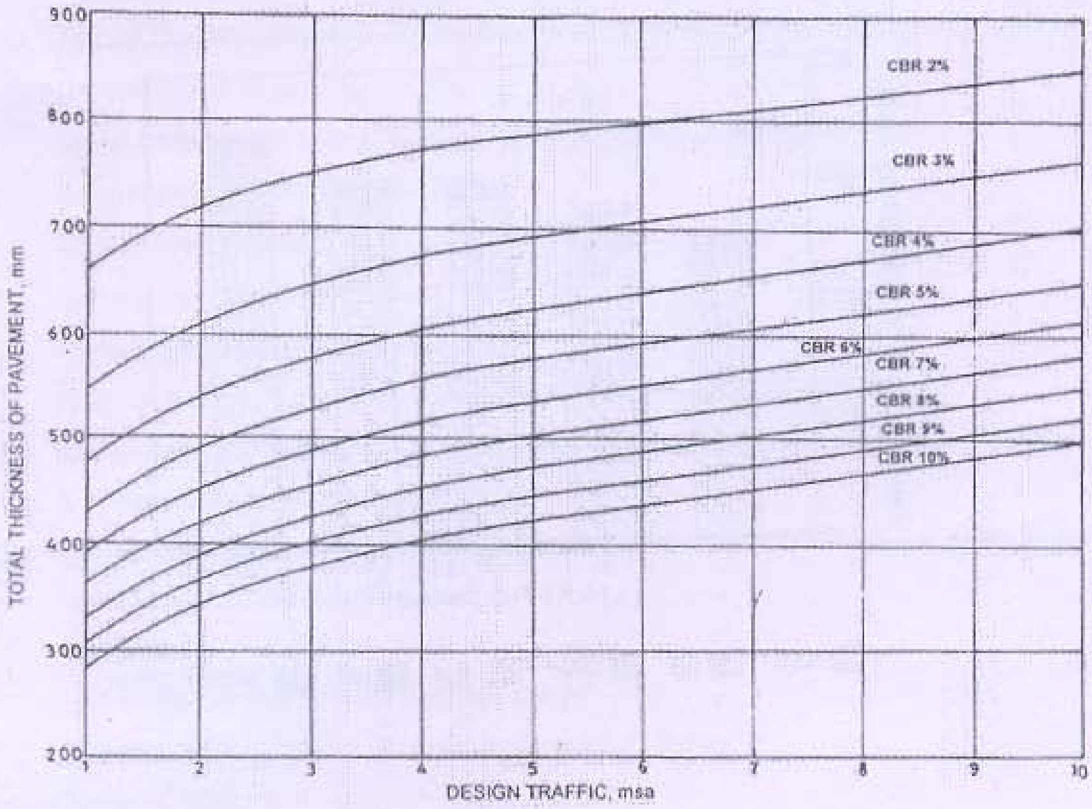

Fig. 1

PAVEMENT DESIGN CATALOGUE

PLATE 1 – RECOMMENDED DESIGNS FOR TRAFFIC RANGE 1-10 msa

CBR 4%					
Cumulative Traffic (msa)	Total Pavement Thickness (mm)	PAVEMENT COMPOSITION			
		Bituminous Surfacing		Granular Base (mm)	Granular Sub-base (mm)
		Wearing Course (mm)	Binder Course (mm)		
1	480	20 PC		225	255
2	540	20 PC	50 BM	225	265
3	580	20 PC	50 BM	250	280
5	620	25 SDBC	60 DBM	250	285
10	700	40 BC	80 DBM	250	330

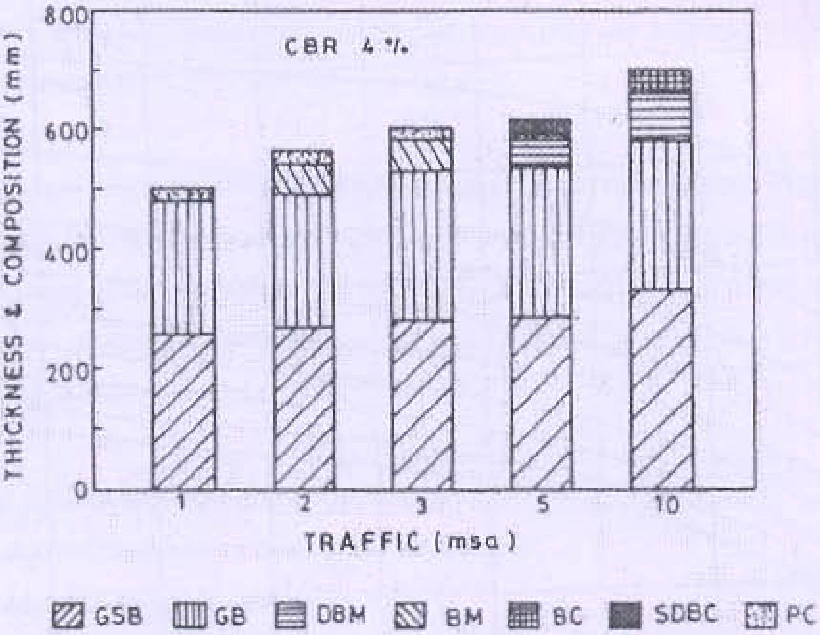

Fig. 2

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